TIPA Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	3D Hierarchical Carbon Host Anode Technology for Na–K Alloys for High-Performance All-Solid-State Batteries			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display (), Secondary Battery (√), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1)]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Electrical and Electronics		Chemicals	
	Medium Category	Small Category	Medium Category	Small Category
	Batteries	Secondary Battery	Polymeric Materials	Materials for Energy/Environmental Industries
Concept of Proposed Technology	• Core concept: ○ Using biomass-derived hard carbon as the primary host to mitigate alloy volume expansion. ○ Introducing a carbon nanotube (CNT) network to create a stand-alone electrode with a conductive framework and mechanical strength. ○ Doping with heteroatoms such as nitrogen (N) and oxygen (O) maximizes wettability with the Na–K alloy, thereby minimizing interfacial resistance. ○ Development of an all-solid-state mixed polymer solid electrolyte including Na-β-Alumina particles • Main function: Realization of a working solid-state battery with high ionic conductivity, chemical and electrochemical stability, and mechanical robustness.			

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	• Intended usage:
	  ○ Enables cost-efficient and energy-saving production of sustainable batteries.
	  ○ Provides high energy and power density with improved material availability.

Rationale for Joint R&D Project	Rationale for Joint R&D Project • The global shift toward renewable energy and electrification requires safe, efficient, and sustainable energy storage systems. • Current lithium-ion technology has reached performance and cost limitations and depends on critical raw materials with limited availability. • There is a strong need to develop next-generation solid-state batteries that combine high energy density, enhanced safety, and low manufacturing costs. • In parallel, alternative chemistries such as sodium-, potassium-, magnesium-, or zinc-based systems are gaining attention for their material abundance and sustainability potential. • Limitations of Existing Technology: Pure Na–K alloy cathodes are prone to leakage due to their liquid state fluidity. Furthermore, dendrite growth and extreme volume changes (> 300%) during charge and discharge compromise the stability of all-solid-state batteries. • Need for a Hybrid Solution: To form an optimal interface with Fraunhofer IAP's and IKTS´s polymer composite electrolyte including Na-beta-Alumina ceramic electrolytes, a highly engineered carbon structure capable of stably containing the alloy while providing an electrical pathway is essential. • The resulting knowledge and materials platform will support industrial partners in diversifying energy storage options and strengthening technological leadership in the global battery market. Necessity for International Collaboration • The development of advanced solid-state batteries requires the integration of multiple scientific disciplines — including polymer chemistry, electrochemistry, materials design, and cell engineering — and thus benefits strongly from international collaboration. • Combining complementary expertise allows the creation of polymer–ceramic hybrid electrolytes with improved ionic conductivity, mechanical stability, and compatibility with different electrodes. • Collaboration between international partners facilitates access to specialized materials synthesis, characterization tools, and industrial upscaling know-how, ensuring the transition from laboratory innovation to practical application. • The cross-border cooperation also fosters standardization efforts and mutual technology transfer.
Key Words	1. All solid state battery 2. Na-K Alloy Anode 3. Hybrid polymer ceramic electrolyte 4. Energy storage

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3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level (4)	End of the Project	Level (7)
R&D Objectives	• Mixed-matrix Na-K electrolyte utilizing solid particles embedded in a conductive polymer • All solid-state Na-K batterie cell • Batterie management system • cost assessment, business plan and commercialization plan			

R&D Target Specifications	• Category Target Specifications Quantitative Target Values
	• Anode Structure: Stand-alone Hybrid Carbon Sheet Thickness: under 100um
	• Alloy Penetration: Na–K alloy filling ratio in the carbon host: over 95%
	• Interfacial Resistance: Electrolyte-Anode interfacial resistance (Rint) : 50Ω per cm 2
	• Lifetime Characteristics: Capacity Retention Rate (at 1,000 cycles) Over 80%
	• Current Density: Reversible Charge/Discharge Current Density: Up to 100 mA/cm2
	• Hybrid polymer–ceramic solid electrolyte for Na–K battery system
	• Electrolyte performance
	  • Ionic conductivity: ≥ 1×10^{-4}S/cm at room temperature.
	  • Electrolyte thickness: 50–200 µm flexible film.
	• Scalable, solution- or melt-based processing.			

R&D Plan	Research approach
	• Develop a universal polymer–ceramic solid electrolyte matrix that can be tailored to Na-K ASSB-systems.
	• Combine materials design, scalable processing and cell integration in iterative development–testing loops.
	• Use model half-cells and full cells to correlate electrolyte properties with electrochemical performance.
	
	Key tasks
	• Task 1 – Materials design and synthesis
	  ○ Selection of suitable Biomass-Based Hard Carbon materials, polymer backbones, salts and ceramic fillers.
	  ○ Synthesis and surface modification of ceramic particles.
	  ○ Formulation of hybrid polymer–ceramic electrolyte compositions.
	• Task 2 – Optimization of Biomass-Based Hard Carbon Synthesis - Doping Technology and Electrolyte processing - characterization
	  ○ Low-cost, Agricultural Byproducts, etc. Selection of biomass feedstock and design of carbonization process.
	  ○ Induction of metal-philic surfaces through N and O doping via plasma or chemical vapor deposition (CVD).			

- Development of thin-film electrolyte processing (casting, coating, lamination).
- Structural, thermal, mechanical and electrochemical characterization (ionic conductivity, stability window, etc.).
- Downselection of lead formulations against the target specifications.

- Task 3 – Fabrication of self-supporting electrodes with CNT networks, Cell integration and electrochemical testing
  - Fabrication of hard carbon-CNT composite films through a binder-free process.
  - Ensuring mechanical flexibility for roll-to-roll compatibility.
  - Assembly of symmetric cells and half-cells to screen interfacial stability and current density limits.
  - Integration into prototype full cells.
  - Cycling tests, rate capability and safety-relevant tests.
- Task 4 – Na–K alloy injection and interface analysis, scaling and evaluation
  - Development of a molten alloy injection process utilizing capillary action.
  - Evaluation of physical and chemical interfacial stability with solid electrolytes.
  - Process optimization for reproducibility and scale-up (e.g. roll-to-roll compatible routes).
  - Techno-economic and sustainability assessment (materials availability, energy demand, cost drivers).
  - Final benchmarking versus state-of-the-art and definition of next development steps.

Methodologies

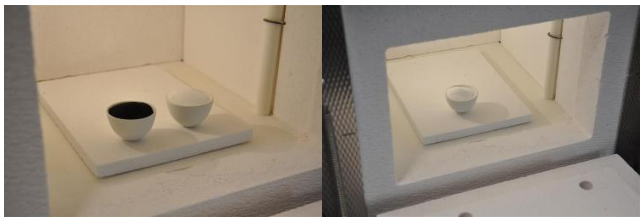
- Polymer and composite processing: solution casting, melt processing, tape/slot-die coating, lamination.
- Materials analysis: DSC/TGA, mechanical testing, microscopy (SEM), XRD, spectroscopy where needed.
- Electrochemical methods: impedance spectroscopy (EIS), cyclic voltammetry, galvanostatic cycling, rate tests, critical current density measurements.

Expected milestones and timeline (example: 3 years)

- Year 1
  - Requirements specification and target values finalized.
  - First electrolyte formulations and ceramic fillers synthesized.
  - Demonstration of Biomass-Based produced Hard Carbon materials
  - Demonstration of thin (50–200 μm) hybrid electrolyte films with target-range ionic conductivity.
  - Initial stability and compatibility tests in symmetric cells.
- Year 2
  - Prototype solid-state cells assembled and tested.
  - Achievement of stable cycling at relevant areal capacities.
- Year 3

	○ Optimized electrolyte formulations and processing route defined. ○ Demonstration of ≥1000 cycles (chemistry- and application-dependent) and validated operation window. ○ Assessment of industrial scalability and preparation of follow-up development/implementation phase.

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	Fraunhofer IAP • Expertise of over 20 years in organic chemistry – synthesis of application dependent customized molecules, monomers and polymers • Development and design of ion conductive polymers • Development of mixed-matrix electrolyte – use of different mixing strategies/ technologies to infiltrate the solid oxide electrolytes in the polymers • Scale up of the polymer synthesis and processing Example of a mixed-matrix polymer electrolyte <u>Fraunhofer IKTS</u> • Development and production of solid oxide electrolyte powder • scale-up experience (solid electrolyte production (50 kg scale)) • active material production • Batterie cell assembling and testing (100 cell testing channels) • Cathode/Anode development solid electrolyte production 
Expected R&R of Korean SME Partner	• Commercialization strategies • Battery / battery management development • Scale-up implementation • Cost assessment and business plan • Smart production • Cathode/Anode production

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5. Expected Outcomes/Impact

Expected Outcomes and Impact	• All-solid-state Na-K battery cell • 3D Hierarchical Carbon Host Anode for Na–K Alloys • mixed-matrix Na-K based conductive solid electrolyte • Operation parameters of the battery cell • First batterie management system for the novel all-solid-state potassium battery • cost assessment and business plan

6. References

(1) Spinning and carbon fiber manufacturing of Cellulose // Boric Acid as A Low-Temperature Graphitization Aid and Its Impact on Structure and Properties of Cellulose-Based Carbon Fibers // https://doi.org/10.3390/polym15214310

(2) Melt spinnable precursor and Carbon fiber from this (ComCarbon technology); 4 Patents claiming precursor material and synthesis, spinning and carbon fiber manufacturing; WO2017084853, WO2017167355, US2017275405, WO2017162268

(3) Patent: Process for the preparation of block copolyquinoxalines with sulfonic or phosphonic acid groups and their use as a proton-conducting membrane; DE102017208805B4

(4) Patent: Block copoly(phenylquinoxalines) with quaternary groups and their use as alkaline conductive membranes; DE102019203597A1

(5) Patent: Sodium ion-conducting membrane based on a sodium beta"-aluminate system, and method for producing the same; DE102016120941A1

(6) Patent: Production of green bodies from Na-β-aluminate; DE102019129869B4

(7) Patent: Electrochemical sodium metal halide battery, and method for producing same; WO2021129905A1

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☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization